

DATA CLEANING CHANGE LOG

Project 1: Data Cleaning & Preparation

DecodeLabs Industrial Training Kit

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Dataset: Dataset for Data Analytics (14 columns)

Total Records: 1,200 rows

Raw File: DecodeLabs_Dataset for Data Analytics.xlsx

Clean File: DecodeLabs_Dataset_Cleaned.xlsx

1. SUMMARY

This change log documents all data cleaning, standardization, and cross-validation steps applied to the Data Analytics dataset (1,200 records, 14 columns). The final dataset meets Project 1's non-negotiable requirements:

- 0% duplicate Order IDs (0 duplicates found in 1,200 records)
- 0% incorrect date formats (ISO 8601: YYYY-MM-DD)
- 309 nulls in CouponCode column replaced with "No Coupon", no other nulls detected
- All cross-validation rules pass or are documented

2. CHANGE LOG

Change ID	Column(s) Affected	Description	Impact	Status
CR001	All 14 columns	Set correct data types (Text, Date, Decimal, Whole Number)	Enabled accurate aggregation, sorting, and date operations across 1,200 records	Resolved
CR002	Order ID	Ran duplicate check on 1,200 records	0 duplicates found	Resolved
CR003	Order ID, Customer ID, Product, Shipping Address, Payment Method, Order Status, Tracking Number, Coupon Code, Referral Source	Trimmed whitespace and cleaned non-printable characters	Standardized text fields across 9 columns	Resolved
CR004	Product, Payment Method, Order Status, Referral Source	Applied Proper Case	Consistent casing across categorical fields	Resolved
CR005	Order ID, Customer ID, Tracking Number, Coupon Code	Applied UPPERCASE	Standardized ID and code formats	Resolved
CR006	Unit Price, Total Price	Converted text to decimal	Enabled numeric calculations and aggregation	Resolved
CR007	Date	Created custom ISO 8601 date column (YYYY-MM-DD)	0% date format errors across 1,200 records	Resolved

CR008	CouponCode	Replaced null and blank values with "No Coupon"	All blank coupon codes labeled; 0 nulls remain	Resolved
CR009	Quantity, Unit Price, Total Price	Added Price_Check cross-validation	0 mismatches	Resolved
CR010	Items in Cart, Quantity	Added IIC_Qty_Check (Items in Cart $\geq$ Quantity)	0 flags	Resolved
CR011	Tracking Number, Order Status	Added Tracking_Flag cross-validation	487 flags identified	Documented
CR012	Quantity, Unit Price, Total Price	Added Qty_Check and Price_Positive_Check	0 errors	Resolved

3. NOTES

➤ Tracking Number vs. Order Status (CR011)

During cross-validation of Tracking Number against Order Status across all records, 487 records (40.6%) were flagged where the tracking number and order status did not match expected rules. These fall into two patterns.

Pattern A: Processing or Pending orders that already have tracking numbers

Pattern B: Cancelled orders that have tracking numbers

What I Did:

- Kept all tracking numbers and status values as-is
- Did not delete or change any records
- Labeled each pattern separately in the Tracking_Flag column: DOCUMENTED for Pattern A, REVIEW_RETURN for Pattern B

Impact:

- 237 records labeled DOCUMENTED
- 250 records labeled REVIEW_RETURN
- No data was changed or removed

Reasoning:

- Chose to document rather than fix because the pattern has no major impact on future analysis. Only shipment/status reports are affected, and analysts can use the tracking number instead of the status.
- Changing the source data would mean guessing what the "real" status should be, which is not my call to make.

4. FINAL VERIFICATION

Requirement	Result	Status
Total Records	1,200	N/A
Duplicate Order IDs	0	PASS
Date Format Errors (non-ISO)	0	PASS
Null Quantity	0	PASS
Null Unit Price	0	PASS
Null Total Price	0	PASS
Null Coupon Code	0	PASS
Negative Quantity	0	PASS
Negative Unit Price	0	PASS
Price Mismatches (Qty × Unit Price)	0	PASS
Cart_Check Flags	0	PASS
Tracking_Check Flags	487	DOCUMENTED
Order ID Format Issues	0	PASS
Customer ID Format Issues	0	PASS

Documented Exceptions: 487 tracking flags (237 DOCUMENTED + 250 REVIEW_RETURN, explained in Notes)